

Homework 4

Problem 1. Explain all the why's in the example of a sequence of differentiable functions converging uniformly to a non-differentiable function given in §3.7 in the textbook.

Define $f_n : [-1, 1] \rightarrow \mathbb{R}$ defined by $f_n(x) = \sqrt{\frac{1}{n^2} + x^2}$. These functions are differentiable. (Why?) We can take the derivative to show that it exists:

$$f_n'(x) = \frac{1}{2} \left(\frac{1}{n^2} + x^2 \right)^{-\frac{1}{2}} \cdot 2x = x \left(\frac{1}{n^2} + x^2 \right)^{-\frac{1}{2}}$$

These functions is defined for all $x \in [-1, 1]$. Thus the derivatives exist for all n .

Also one can check that

$$|x| \leq f_n(x) \leq |x| + \frac{1}{n}$$

for all $x \in [-1, 1]$. (Why?) Its clear that we have

$$f_n(x) = \sqrt{\frac{1}{n^2} + x^2} \geq \sqrt{x^2} = |x|$$

since $\frac{1}{n^2}$ is just a positive number. For the second inequaltiy we have

$$\begin{aligned} f_n(x) &= \sqrt{\frac{1}{n^2} + x^2} \\ \Leftrightarrow (f_n(x))^2 &= \frac{1}{n^2} + x^2 \leq \frac{1}{n^2} + 2\frac{1}{n} |x| + x^2 = \left(\frac{1}{n} + |x| \right)^2 \\ \Leftrightarrow f_n(x) &\leq \frac{1}{n} + |x|. \end{aligned}$$

Thus we have

$$|x| \leq f_n(x) \leq \frac{1}{n} + |x|.$$

So by the squeeze theorem $f_n(x) \rightarrow f(x) := |x|$ uniformly. But this function is not differentiable at $x = 0$ (why?). Note how $|x|$ is defined:

$$|x| := \begin{cases} -x & \text{if } x < 0 \\ x & \text{if } x > 0 \end{cases}.$$

We see that as x approaches 0 from the left, the derivative is -1 , but from the right the derivative is 1. Hence there is a mismatch in the limits from both sides which implies that such a limit does not exist. We conclude that the uniform limit of a differentiable function need not be differentiable.

Problem 2. Let $[a, b]$ be an interval, and for every integer $n \geq 1$, let $f_n : [a, b] \rightarrow \mathbb{R}$ be a differentiable function whose derivative $f_n' : [a, b] \rightarrow \mathbb{R}$ is continuous. Suppose that the derivatives f_n' converge uniformly to a function $g : [a, b] \rightarrow \mathbb{R}$. Suppose that there also exists a point x_0 such that the limit $\lim_{n \rightarrow \infty} f_n(x_0)$ exists. Then the functions f_n converge uniformly to a differentiable function f , and the derivative of f equals g .

Proof. Since f_n' continuous, then by FTC we have $f_n(x) - f_n(x_0) = \int_{[x_0, x]} f_n'$ when $x \in [x_0, x]$, and $f_n(x) - f_n(x_0) = -\int_{[x, x_0]} f_n'$ when $x \in [x, x_0]$. Let $L := \lim_{n \rightarrow \infty} f_n(x_0)$. By the hypothesis, L exists. Since each f_n' is continuous and $f_n' \rightarrow g$ uniformly, then g is also continuous. Now define $f(x) := L + \int_{[x_0, x]} g$, for all $x \in [a, b]$. Let $\varepsilon > 0$. Then by the uniform convergence of f_n' to g , we have there exists an N_1 such that for all $n > N_1, x \in [a, b]$, $|f_n'(x) - g(x)| < \varepsilon$. Since $f_n(x_0)$ converges to L , then we have that there exists N_2 such that for all $n > N_2, |f_n(x_0) - L| < \varepsilon$. Then we have,

$$\begin{aligned} \left| \int_{[x_0, x]} (f_n' - g) \right| &\stackrel{(\Delta)}{\leq} \int_{[x_0, x]} |f_n' - g| \\ &< \int_{[x_0, x]} \varepsilon \, dt \\ &= \varepsilon |x - x_0|. \end{aligned}$$

Note that $|x - x_0| \leq (b - a)$ since $x, x_0 \in [a, b]$. So we get,

$$\left| \int_{[x_0, x]} (f_n' - g) \right| < \varepsilon(b - a).$$

Take $N := \max(N_1, N_2)$. Then we have for all $n > N, x \in [a, b]$,

$$\begin{aligned} |f_n(x) - f(x)| &= \left| f_n(x_0) + \int_{[x_0, x]} f_n' - L - \int_{[x_0, x]} g \right| \\ &\leq |f_n(x_0) - L| + \left| \int_{[x_0, x]} (f_n' - g) \right| \\ &< \varepsilon + \varepsilon(b - a) = \varepsilon(b - a + 1). \end{aligned}$$

Since ε was taken to be arbitrary, then we have that

$$|f_n(x) - f(x)| \rightarrow 0$$

Thus, $f_n \rightarrow f(x)$ uniformly. It remains to show that we have $f'(x) = g$. We can simply take the derivative of f : $f'(x) = \left(L + \int_{[x_0, x]} g \right)' \stackrel{\text{FTC}}{=} 0 + g(x) = g(x)$. □

Problem 3. Suppose $f_n : [a, b] \rightarrow \mathbb{R}$ is differentiable and $f'_n : [a, b] \rightarrow \mathbb{R}$ is continuous for every n . If $\sum_{n=1}^{\infty} \|f'_n\|_{\infty}$ is absolutely convergent, and $\sum_{n=1}^{\infty} f(x_0)$ converges for some $x_0 \in [a, b]$, then $\sum f_n$ converges uniformly to some differentiable function on $[a, b]$, and

$$\frac{d}{dx} \sum_{n=1}^{\infty} f_n(x) = \sum_{n=1}^{\infty} \frac{d}{dx} f_n(x) = \sum_{n=1}^{\infty} f'_n(x)$$

Proof. Since all f_n is differentiable, then a finite sum $\sum_{n=1}^N f_n$ is also differentiable for every N . Similarly, since all f'_n are continuous, then $\sum_{n=1}^N f'_n$ also is continuous. Additionally by assumption since $\sum_{n=1}^{\infty} \|f'_n\|_{\infty}$ converges, then by the Weierstrass M-Test $\sum_{n=1}^N f'_n$ converges uniformly to $\sum_{n=1}^{\infty} f'_n$. By assumption we have $\sum_{n=1}^{\infty} f_n(x_0)$ exists for some $x_0 \in [a, b]$. Then $\sum_{n=1}^N f_n$ and $\sum_{n=1}^N f'_n$ satisfy the hypothesis of Theorem 3.7.1. So we have that $\sum_{n=1}^N f_n$ converges uniformly to $\sum_{n=1}^{\infty} f_n$ which is differentiable, and

$$\frac{d}{dx} \sum_{n=1}^{\infty} f_n = \sum_{n=1}^{\infty} f'_n = \sum_{n=1}^{\infty} \frac{d}{dx} f_n$$

This is what we wanted to show. □

Problem 4. Show that $\sum_{n=1}^{\infty} 4^{-n} \cos(32^n \pi x)$ converges uniformly.

Proof. Note, $\sum_{n=1}^{\infty} \|4^{-n} \cos(32^n \pi x)\|_{\infty} = \sum_{n=1}^{\infty} 4^{-n}$ since $|\cos(32^n \pi x)| \leq 1$. We know that $\sum_{n=1}^{\infty} 4^{-n}$ converges, so by Weierstrass M-Test we have that $\sum_{n=1}^{\infty} 4^{-n} \cos(32^n \pi x)$ converges uniformly. □

Problem 5. Let $\sum_{n=0}^{\infty} C_n(x-a)^n$ be a formal power series, and let R be its radius of convergence.

- (a) If $x \in \mathbb{R}$ is such that $|x-a| > R$, then $\sum_{n=0}^{\infty} C_n(x-a)^n$ is divergent for that value of x .
 (b) If $|x-a| < R$, then $\sum_{n=0}^{\infty} C_n(x-a)^n$ converges absolutely.

Proof of (a). Note, $\limsup |C_n(x-a)^n|^{\frac{1}{n}} = \limsup |C_n|^{\frac{1}{n}} |(x-a)^n|^{\frac{1}{n}} = \limsup |C_n|^{\frac{1}{n}} |x-a| > \limsup |C_n|^{\frac{1}{n}} R$ since $|x-a| > R$ by assumption. Then we have

$$\limsup |C_n|^{\frac{1}{n}} R = \limsup |C_n|^{\frac{1}{n}} \frac{1}{\limsup |C_n|^{\frac{1}{n}} R} = 1.$$

So we have that $\limsup |C_n(x-a)^n|^{\frac{1}{n}} > 1$. So by the root test, $\sum_{n=0}^{\infty} C_n(x-a)^n$ diverges. □

Proof of (b). A similar argument to (a) can be applied here. We have

$$\limsup |C_n(x-a)^n|^{\frac{1}{n}} < \limsup |C_n|^{\frac{1}{n}} R = \limsup |C_n|^{\frac{1}{n}} \frac{1}{\limsup |C_n|^{\frac{1}{n}}} = 1.$$

By the root test, $\sum_{n=1}^{\infty} C_n(x-a)^n$ converges absolutely. $\square$

Problem 6. Now assume $R > 0$. Let $f : (a-R, a+R) \rightarrow \mathbb{R}$ be defined by $f(x) := \sum_{n=1}^{\infty} C_n(x-a)^n$ (which exists by (b)).

(c) For all $0 < r < R$, $\sum_{n=1}^{\infty} C_n(x-a)^n$ converges uniformly to f on $[a-r, a+r]$. In particular, f is continuous on $(a-R, a+R)$.

(d) f is differentiable on $(a-R, a+R)$, and $\sum_{n=1}^{\infty} nC_n(x-a)^{n-1}$ converges uniformly to f' on $[a-r, a+r]$ for every $0 < r < R$.

(e) For any closed interval $[y, z] \subset (a-R, a+R)$,

$$\int_{[y,z]} f = \sum_{n=1}^{\infty} C_n \frac{(z-a)^{n+1} - (y-a)^{n+1}}{n+1}.$$

Proof of (c). Let $f_n : [a-r, a+r] \rightarrow \mathbb{R}$ be defined by $f_n(x) := C_n(x-a)^n$. Then

$$\|f_n\|_{\infty} = \sup\{|C_n(x-a)^n| \mid x \in [a-r, a+r]\} = |C_n|r^n.$$

Since $|(a+r)-a| = r < R$, then by (b) $\sum C_n((a+r)-a)^n = \sum C_n r^n$ converges. So by Weierstrass M-Test, $\sum f_n(x) = \sum C_n(x-a)^n$ converges uniformly. $\square$

Proof of (d). Let $f_{n(x)} = C_n(x-a)^n$. Then $f'_n(x) = nC_n(x-a)^{n-1}$. Note, $\sum_{n=1}^{\infty} f'_n(x) = \sum_{n=0}^{\infty} nC_n(x-a)^{n-1} = \frac{1}{x-a} \sum_{n=1}^{\infty} nC_n(x-a)^n$. We can consider all $x \neq a$ since at $x = a$, the series is just 0. Note that $\lim |n|^{\frac{1}{n}} = 1$. So observe that

$$\frac{1}{\limsup |nC_n|^{\frac{1}{n}}} = \frac{1}{\lim |n|^{\frac{1}{n}}} \frac{1}{\limsup |C_n|^{\frac{1}{n}}} = R.$$

So $\sum_{n=1}^{\infty} nC_n(x-a)^n$ has radius of convergence R . Since for fixed $x \neq a$, $\frac{1}{x-a}$ doesn't affect convergence then $\sum_{n=1}^{\infty} C_n(x-a)^{n-1}$ has radius of convergence R . By (c), $\sum_{n=1}^{\infty} nC_n(x-a)^{n-1}$ converges uniformly. Note that taking $x = a \in [a-r, a+r]$, we have $\sum_{n=1}^{\infty} C_n(x-a)^n = 0$. So by Theorem 3.7.3 we have $f'(x) = \sum_{n=1}^{\infty} f'_n(x)$ $\square$

Proof of (e). By (c), $\sum_{n=0}^{\infty} C_n(x-a)^n$ converges uniformly for all $[a-r, a+r]$. Then for all intervals $[y, z] \subset (a-R, a+R)$, there exists an $r > 0$ such that $[y, z] \subset [a-r, a+r]$. So $\sum_{n=0}^{\infty} C_n(x-a)^n$ converges uniformly on $[y, z]$. Then we have

$$\begin{aligned}
\int_{[y,z]} \sum_{n=0}^{\infty} C_n (x-a)^n &= \sum_{n=0}^{\infty} \int_{[y,z]} C_n (x-a)^n \\
&= \sum_{n=0}^{\infty} C_n \left[\frac{(x-a)^{n+1}}{n+1} \right]_y^z \\
&= \sum_{n=0}^{\infty} C_n \frac{(z-a)^{n+1} - (y-a)^{n+1}}{n+1}
\end{aligned}$$

□

Problem 7. Let $a \in \text{int}E$, $E \subseteq \mathbb{R}$, and f a real analytic function at a . So $f(x) = \sum_{n=0}^{\infty} C_n (x-a)^n$ on $(a-r, a+r)$ for some $r > 0$. Then for any $k \geq 0$, f is k -times differentiable on $(a-r, a+r)$ and

$$f^{(k)}(x) = \sum_{n=0}^{\infty} C_{n+k} (n+1)(n+1) \cdots (n+k) (x-a)^n = \sum_{n=0}^{\infty} C_{n+k} \frac{(n+k)!}{n!} (x-a)^n$$

on $(a-r, a+r)$.

Proof. We will proceed by induction on k . For $k = 0$ we have $f^{(0)}(x) = f(x) = \sum_{n=0}^{\infty} C_{n+0} \frac{(n+0)!}{n!} (x-a)^n$. Now assume we have the k -th derivative such that $f^{(k)}(x) = \sum_{n=0}^{\infty} C_{n+k} \frac{(n+k)!}{n!} (x-a)^n$. Then $f^{(k)}$ is real analytic and hence uniformly convergent. So by Theorem 4.1.6 (d) we have,

$$\begin{aligned}
f^{(k+1)}(x) &= \sum_{n=0}^{\infty} C_{n+k} \frac{(n+k)!}{n!} n (x-a)^{n-1} \\
&= \sum_{n=1}^{\infty} C_{n+k} \frac{(n+k)!}{n!} n (x-a)^{n-1} \\
&= \sum_{n=0}^{\infty} C_{n+k+1} \frac{(n+k+1)!}{(n+1)!} (n+1) (x-a)^n \\
&= \sum_{n=0}^{\infty} C_{n+(k+1)} \frac{(n+(k+1))!}{n!} (x-a)^n.
\end{aligned}$$

By induction on k , we are done.

□